Bootstrap Notes

Bootstrap 5 (with Bootstrap 3 components)

Student Study Notes | Syntax + Explanation + Examples

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1. Introduction to Bootstrap

Bootstrap is the most popular free **front-end framework** for building responsive, mobile-first websites quickly. It provides ready-made CSS classes and JavaScript components (buttons, navbars, grids, modals, carousels).

- Created by Twitter developers (Mark Otto and Jacob Thornton), released in 2011.
- Versions: Bootstrap 3 (panels, wells, labels, pager), Bootstrap 4, **Bootstrap 5 (latest; no jQuery needed)**.
- **Advantages**: saves time, responsive by default, consistent design, works in all modern browsers, large community.
- Two ways to use it: **CDN link** (internet needed) or **download** the files and host locally.

> **Note:** In these notes, examples use Bootstrap 5. Panels, Wells, Labels and Pager exist in Bootstrap 3, so their Bootstrap 3 code is given along with the Bootstrap 5 replacement.

2. Bootstrap Basics (Setup and Template)

Starter template using CDN

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Bootstrap Page</title>
  <!-- Bootstrap 5 CSS -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
        rel="stylesheet">
</head>
<body>
  <div class="container">
    <h1 class="text-primary">Hello Bootstrap!</h1>
    <button class="btn btn-success">Click me</button>
  </div>

  <!-- Bootstrap 5 JS bundle (includes Popper) -->
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"><
/script>
</body>
</html>
```

Containers

Class	Description
`.container`	Fixed width that changes at each breakpoint, centered.
`.container-fluid`	Full width of the screen always.
`.container-md etc.`	100% wide until the md breakpoint, then fixed.

- Always include the **viewport meta tag** and the **HTML5 doctype**.
- Bootstrap uses a reboot (normalize) stylesheet so all browsers look the same.

3. Bootstrap Grid System

The grid is based on **12 columns** per row and uses **flexbox**. Structure: **container > row > col**. Column numbers in a row should add up to 12 (extra columns wrap to the next line).

Prefix	Breakpoint (screen width)
col-	Extra small: less than 576px (all sizes)
col-sm-	Small: 576px and up
col-md-	Medium: 768px and up
col-lg-	Large: 992px and up
col-xl-	Extra large: 1200px and up
col-xxl-	Extra extra large: 1400px and up

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Column 1 (4/12)</div>
    <div class="col-md-8">Column 2 (8/12)</div>
  </div>

  <!-- Equal width columns -->
  <div class="row">
    <div class="col">A</div><div class="col">B</div><div class="col">C</div>
  </div>

  <!-- Responsive: stack on mobile, 2 on tablet, 4 on desktop -->
  <div class="row">
    <div class="col-12 col-sm-6 col-lg-3">Box</div>
    <div class="col-12 col-sm-6 col-lg-3">Box</div>
    <div class="col-12 col-sm-6 col-lg-3">Box</div>
    <div class="col-12 col-sm-6 col-lg-3">Box</div>
  </div>
</div>
```

Useful grid classes

- **offset-md-2** : pushes a column to the right by 2 columns.
- **g-3** : gutter (gap) between columns.
- **justify-content-center / align-items-center** : alignment inside a row.
- **order-1, order-2** : change column order.

4. Bootstrap CSS (Typography, Tables, Forms, Buttons, Images, Utilities)

Typography

```
<h1 class="display-4">Big heading</h1>
<p class="lead">Lead paragraph</p>
<p class="text-center text-uppercase fw-bold text-danger">Styled text</p>
<mark>highlight</mark>  <small>small</small>
<blockquote class="blockquote">Quote text</blockquote>
```

Colours and utility classes

Class	Meaning
text-primary / success / danger / warning / info	Text colours.
bg-primary / bg-light / bg-dark	Background colours.
m-3, mt-2, mx-auto, p-4	Margin and padding (0 to 5).
d-none, d-md-block	Show/hide at breakpoints.
d-flex, justify-content-between	Flexbox helpers.
rounded, border, shadow	Border radius, border, shadow.
w-50, h-100	Width / height %.

Tables

```
<table class="table table-striped table-bordered table-hover">
  <thead class="table-dark">
    <tr><th>Name</th><th>City</th></tr>
  </thead>
  <tbody>
    <tr><td>Ravi</td><td>Mathura</td></tr>
    <tr><td>Anita</td><td>Agra</td></tr>
  </tbody>
</table>
<!-- wrap in <div class="table-responsive"> for horizontal scroll on mobile -->
```

Buttons

```
<button class="btn btn-primary">Primary</button>
<button class="btn btn-outline-danger btn-lg">Large Outline</button>
<button class="btn btn-secondary btn-sm">Small</button>
<button class="btn btn-success w-100">Block button</button>
<div class="btn-group"><button class="btn btn-dark">A</button><button class="btn
btn-dark">B</button></div>
```

Forms

```
<form>
  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" class="form-control" id="email"
placeholder="name@example.com">
  </div>
  <div class="mb-3">
    <label class="form-label">City</label>
    <select class="form-select"><option>Mathura</option><option>Agra</option></
select>
  </div>
  <div class="form-check">
    <input class="form-check-input" type="checkbox" id="c1">
    <label class="form-check-label" for="c1">Remember me</label>
  </div>
  <button type="submit" class="btn btn-primary mt-3">Submit</button>
</form>
```

Images

```
<img src="a.jpg" class="img-fluid" alt="Responsive">
<img src="a.jpg" class="img-thumbnail" alt="Thumbnail">
<img src="a.jpg" class="rounded-circle" width="100" alt="Circle">
```

5. Bootstrap JS (JavaScript Components)

Bootstrap JavaScript adds interactive behaviour using **data attributes** (no JS coding needed). Include **bootstrap.bundle.min.js** (it contains Popper). Bootstrap 5 does not need jQuery.

Component	Purpose
`Modal`	Popup dialog box.
`Dropdown`	Toggleable menu.
`Collapse / Accordion`	Show/hide content.
`Tooltip / Popover`	Small hover / click info boxes.
`Carousel`	Image slider.
`Offcanvas / Toast / Scrollspy / Tabs`	Side panel, notification, scroll tracking, tab panels.

Modal

```
<button class="btn btn-primary" data-bs-toggle="modal" data-bs-
target="#myModal">Open</button>

<div class="modal fade" id="myModal" tabindex="-1">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <h5 class="modal-title">Title</h5>
        <button class="btn-close" data-bs-dismiss="modal"></button>
      </div>
      <div class="modal-body">Modal body text.</div>
      <div class="modal-footer">
        <button class="btn btn-secondary" data-bs-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```

Dropdown, Collapse, Tooltip

```
<div class="dropdown">
  <button class="btn btn-info dropdown-toggle"
data-bs-toggle="dropdown">Menu</button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another</a></li>
  </ul>
</div>

<button class="btn btn-dark" data-bs-toggle="collapse" data-bs-
target="#info">Toggle</button>
<div id="info" class="collapse">Hidden content appears here.</div>
```

```
<button class="btn btn-light" data-bs-toggle="tooltip" title="Hello!">Hover
me</button>
<script>
  document.querySelectorAll('[data-bs-toggle="tooltip"]')
          .forEach(el => new bootstrap.Tooltip(el));   /* tooltips must be
initialised */
</script>
```

Navbar (responsive)

```
<nav class="navbar navbar-expand-md navbar-dark bg-dark">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">MySite</a>
    <button class="navbar-toggler" data-bs-toggle="collapse" data-bs-target="#nav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="nav">
      <ul class="navbar-nav">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">About</a></li>
      </ul>
    </div>
  </div>
</nav>
```

6. Bootstrap Themes

A **theme** changes the overall look of Bootstrap (colours, fonts, shapes) without changing the HTML.

- **Bootswatch** (bootswatch.com): free ready-made themes (Cerulean, Darkly, Flatly, Lux, Sandstone…). Just replace the CSS link with the theme CSS link.
- **Official Bootstrap Themes** marketplace: paid professional templates and dashboards.
- **Customising with Sass**: change variables like $primary before compiling.
- **CSS variables** (Bootstrap 5.3): override at runtime.
- **Dark mode** (5.3): add data-bs-theme="dark" to html or any element.

```
<!-- Using a Bootswatch theme: replace Bootstrap CSS link -->
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootswatch@5.3.3/dist/darkly/bootstrap.min.css">

<!-- Own override file loaded after Bootstrap -->
<link rel="stylesheet" href="custom.css">

/* custom.css */
:root { --bs-primary: #6f42c1; }
.btn-primary { background-color: #6f42c1; border-color: #6f42c1; }

<!-- Dark mode -->
<html lang="en" data-bs-theme="dark">
```

7. Alerts and Wells

Alerts (Bootstrap 5)

Alerts display feedback messages. Use contextual classes: **alert-success, alert-info, alert-warning, alert-danger, alert-primary, alert-secondary, alert-light, alert-dark**.

```
<div class="alert alert-success">
  <strong>Success!</strong> Your form was submitted.
</div>

<div class="alert alert-danger alert-dismissible fade show">
  <strong>Error!</strong> Something went wrong.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>

<div class="alert alert-info">Read the <a href="#"
class="alert-link">guide</a>.</div>
```

Wells (Bootstrap 3 only)

A **well** is a grey inset box with a rounded border, used to highlight content. Sizes: well-sm and well-lg. It was removed in Bootstrap 4/5.

```
<!-- Bootstrap 3 -->
<div class="well">Basic well</div>
<div class="well well-lg">Large well</div>
<div class="well well-sm">Small well</div>

<!-- Bootstrap 5 replacement -->
<div class="p-3 bg-light border rounded">Looks like a well</div>
```

8. Badges and Labels

Badges (Bootstrap 5)

A **badge** is a small counter or label, e.g. number of notifications.

```
<h1>Heading <span class="badge bg-secondary">New</span></h1>
<button class="btn btn-primary">
  Messages <span class="badge bg-light text-dark">4</span>
</button>
<span class="badge rounded-pill bg-danger">99+</span>
<span class="badge bg-success">Success</span>
```

Labels (Bootstrap 3 only)

In Bootstrap 3, **labels** show small tags: label-default, label-primary, label-success, label-info, label-warning, label-danger. In Bootstrap 4/5 they were replaced by badges.

```
<!-- Bootstrap 3 -->
<h3>Example <span class="label label-success">New</span></h3>
<span class="label label-danger">Danger</span>
<span class="badge">42</span>        <!-- BS3 badge -->
```

9. Panels (Bootstrap 3) and Cards (Bootstrap 5)

Panels are bordered boxes with optional header and footer. They exist in Bootstrap 3. In Bootstrap 4/5 panels were replaced by **cards**.

```
<!-- Bootstrap 3 -->
<div class="panel panel-primary">
  <div class="panel-heading">Panel Heading</div>
  <div class="panel-body">Panel content</div>
  <div class="panel-footer">Panel Footer</div>
</div>
<!-- Contextual: panel-default, panel-primary, panel-success, panel-info, panel-
warning, panel-danger -->
```

```
<!-- Bootstrap 5 Card -->
<div class="card" style="width:18rem;">
  <img src="pic.jpg" class="card-img-top" alt="...">
  <div class="card-header">Header</div>
  <div class="card-body">
    <h5 class="card-title">Card title</h5>
    <p class="card-text">Some text.</p>
    <a href="#" class="btn btn-primary">Go</a>
  </div>
  <div class="card-footer text-muted">Footer</div>
</div>
```

- Panel groups (accordion) in BS3 use panel-group + collapse; in BS5 use the **accordion** component.

10. Pagination

Pagination divides long content into numbered pages with links.

```
<nav aria-label="Page navigation">
  <ul class="pagination">
    <li class="page-item disabled"><a class="page-link" href="#">Previous</a></li>
    <li class="page-item active"><a class="page-link" href="#">1</a></li>
    <li class="page-item"><a class="page-link" href="#">2</a></li>
    <li class="page-item"><a class="page-link" href="#">3</a></li>
    <li class="page-item"><a class="page-link" href="#">Next</a></li>
  </ul>
</nav>

<!-- Sizes and alignment -->
<ul class="pagination pagination-lg justify-content-center"> ... </ul>
```

Class	Use
.pagination	List wrapper.
.page-item / .page-link	Each item / link.
.active	Current page.
.disabled	Unavailable link.
.pagination-lg / -sm	Larger / smaller size.
.justify-content-center	Align pagination (BS4/5).

Note: In Bootstrap 3 the markup is <ul class="pagination">
<a>.. without page-item/page-link; use .active and .disabled on the <li>.

11. Pager (Bootstrap 3)

A **pager** gives simple **Previous / Next** links, useful for blogs or articles. It exists in Bootstrap 3. In Bootstrap 4/5 use pagination with only two items.

```
<!-- Bootstrap 3 -->
<ul class="pager">
  <li><a href="#">Previous</a></li>
  <li><a href="#">Next</a></li>
</ul>

<!-- Aligned to the edges -->
<ul class="pager">
  <li class="previous"><a href="#">&larr; Older</a></li>
  <li class="next"><a href="#">Newer &rarr;</a></li>
</ul>

<!-- Bootstrap 5 alternative -->
<nav>
  <ul class="pagination justify-content-between">
    <li class="page-item"><a class="page-link" href="#">Previous</a></li>
    <li class="page-item"><a class="page-link" href="#">Next</a></li>
  </ul>
</nav>
```

- Use class **disabled** on a <li> to disable a link.

12. Carousel

A **carousel** is a slideshow that cycles through images or text slides.

```
<div id="demo" class="carousel slide" data-bs-ride="carousel">

  <!-- Indicators -->
  <div class="carousel-indicators">
    <button data-bs-target="#demo" data-bs-slide-to="0" class="active"></button>
    <button data-bs-target="#demo" data-bs-slide-to="1"></button>
  </div>

  <!-- Slides -->
  <div class="carousel-inner">
    <div class="carousel-item active">
      <img src="img1.jpg" class="d-block w-100" alt="First">
      <div class="carousel-caption"><h5>First slide</h5><p>Caption</p></div>
    </div>
    <div class="carousel-item">
      <img src="img2.jpg" class="d-block w-100" alt="Second">
    </div>
  </div>

  <!-- Controls -->
  <button class="carousel-control-prev" data-bs-target="#demo" data-bs-slide="prev">
    <span class="carousel-control-prev-icon"></span>
  </button>
  <button class="carousel-control-next" data-bs-target="#demo" data-bs-slide="next">
    <span class="carousel-control-next-icon"></span>
  </button>
</div>
```

- The first item must have the class **active**.
- **carousel-fade** gives a fade effect; **data-bs-interval="3000"** sets the time per slide.
- In Bootstrap 3: data-ride="carousel", .item, .carousel-control left/right, data-slide.

13. Progress Bar

A **progress bar** shows the completion status of a task.

```
<!-- Basic -->
<div class="progress">
  <div class="progress-bar" style="width:70%">70%</div>
</div>

<!-- Coloured, striped and animated -->
<div class="progress" style="height:25px;">
  <div class="progress-bar bg-success progress-bar-striped progress-bar-animated"
       role="progressbar" style="width:60%" aria-valuenow="60" aria-valuemin="0"
aria-valuemax="100">
    60%
  </div>
</div>

<!-- Stacked bars -->
<div class="progress">
  <div class="progress-bar bg-success" style="width:40%">Free</div>
  <div class="progress-bar bg-warning" style="width:30%">Warning</div>
  <div class="progress-bar bg-danger"  style="width:20%">Danger</div>
</div>
```

Class	Use
.progress	Outer container (set height here).
.progress-bar	The filled bar; width sets the value.
.bg-success / info / warning / danger	Colour.
.progress-bar-striped	Striped look.
.progress-bar-animated	Moving stripes (BS3: .active).